25 Structured Exam-Style Questions **Name:** _____ **Date:** _____ **Score:** _____ / 25

Systems of Linear ODEs & Phase Plane Topology Practice Exercises

1 MCQ

The matrix of $\mathbf{x}' = A\mathbf{x}$ has eigenvalues $\pm i$. The origin is a:

A saddle point (unstable)
B unstable spiral
C stable node
D center (stable, not asymptotically)

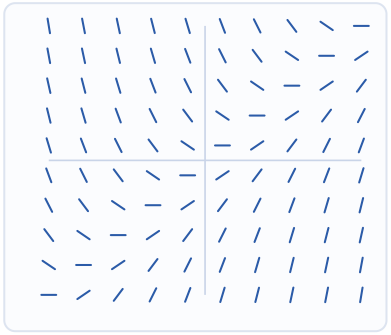
2 MCQ

Solve $x' = 2x,\ y' = -y$ with $x(0) = -1,\ y(0) = 3$. Find $x(\ln(2))$.

A $-\frac{1}{2}$
B 4
C 12
D -4

3 MCQ

With $x_1 = y, x_2 = y'$, the equation $y'' + 3y = 0$ becomes $\mathbf{x}' = A\mathbf{x}$. Find A.



A $\begin{bmatrix} 0 & 1 \\ 3 & 0 \end{bmatrix}$
B $\begin{bmatrix} 0 & 1 \\ -3 & 0 \end{bmatrix}$
C $\begin{bmatrix} 1 & 0 \\ -3 & 0 \end{bmatrix}$
D $\begin{bmatrix} 0 & 1 \\ 0 & -3 \end{bmatrix}$

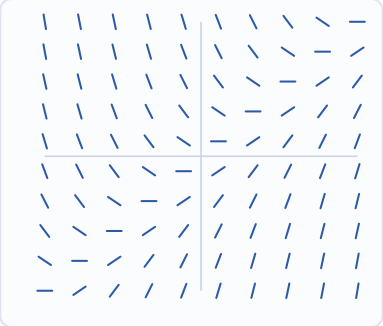
4 MCQ

The matrix of $\mathbf{x}' = A\mathbf{x}$ has eigenvalues $\pm 5i$. The origin is a:

A center (stable, not asymptotically)
B unstable node
C saddle point (unstable)
D unstable spiral

5 MCQ

Solve $x' = -x,\ y' = -y$ with $x(0) = -2,\ y(0) = 4$. Find $y(\ln(2))$.



A -2

B 2

C -1

D 4

6 MCQ

With $x_1 = y, x_2 = y'$, the equation $y'' + y' + y = 0$ becomes $\mathbf{x}' = A\mathbf{x}$. Find A.

A $\begin{bmatrix} 0 & 1 \\ -1 & -1 \end{bmatrix}$

B $\begin{bmatrix} 0 & 1 \\ 1 & 1 \end{bmatrix}$

C $\begin{bmatrix} 1 & 0 \\ -1 & -1 \end{bmatrix}$

D $\begin{bmatrix} -1 & -1 \\ 1 & 0 \end{bmatrix}$

7 MCQ

The matrix of $\mathbf{x}' = A\mathbf{x}$ has eigenvalues $-2 \pm 4i$. The origin is a:

A stable spiral

B unstable spiral

C unstable node

D stable node

8 MCQ

Solve $x' = -x,\ y' = y$ with $x(0) = 4,\ y(0) = -1$. Find $x(1)$.

A $4e$

B $-\frac{4}{e}$

C $-\frac{1}{e}$

D $\frac{4}{e}$

9 MCQ

The second-order linear homogeneous ODE $u''(s) - 40u'(s) + 144u(s) = 0$ has characteristic roots λ_1, λ_2. Determine the trace sum $\lambda_1 + \lambda_2$.

A 44

B 43

C 40

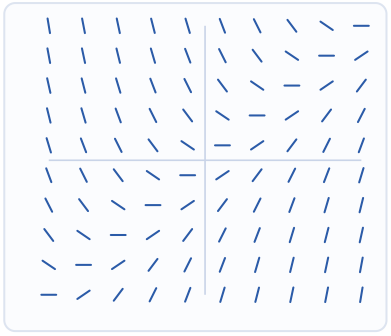
D 24

10 MCQ

Consider the system of linear differential equations given by:

$$\begin{cases} \dfrac{dx}{dt} = -x + 2y \\ \dfrac{dy}{dt} = 3x - 2y \end{cases}$$

Let the general solution be of the form $\mathbf{x}(t) = c_1\mathbf{v}_1 e^{\lambda_1 t} + c_2\mathbf{v}_2 e^{\lambda_2 t}$, where $\lambda_1 < \lambda_2$. Determine the value of λ_2.



A 2

B 1

C 3

D 4

11 MCQ

The separable initial value problem $\dfrac{du}{ds} = 18s$ satisfies $u(0) = 8$. Determine the solution value $u(7)$.

A 452

B 458

C 449

D 441

12 MCQ

Consider the system of linear differential equations: $\begin{aligned} \dfrac{dx}{dt} &= ax + by \\ \dfrac{dy}{dt} &= cx + dy \end{aligned}$ where a, b, c, d are constants. The characteristic equation for the system is $\lambda^2 - (a+d)\lambda + (ad - bc) = 0$. We are given that the eigenvalues of the system are $\lambda_1 = 2$ and $\lambda_2 = -1$. The trace of the coefficient matrix, $T = a + d$, is 1. Determine the value of the determinant of the coefficient matrix, $D = ad - bc$.

A -2

B -4

C 2

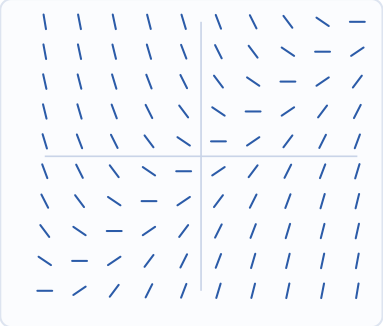
D 4

13 MCQ

Consider the system of linear differential equations given by:

$$\frac{dx}{dt} = x + 2y$$
$$\frac{dy}{dt} = 3x + 2y$$

If the solution to this system is of the form $x(t) = c_1 e^{\lambda_1 t} + c_2 e^{\lambda_2 t}$ and $y(t) = k_1 c_1 e^{\lambda_1 t} + k_2 c_2 e^{\lambda_2 t}$, what is the value of the larger eigenvalue λ?



A 3

B 6

C 5

D 4

14 MCQ

With $x_1 = y$, $x_2 = y'$, the equation $y'' + 3y' - 4y = 0$ becomes $\mathbf{x}' = A\mathbf{x}$. Find A.

A $\begin{bmatrix} 1 & 0 \\ 4 & -3 \end{bmatrix}$

B $\begin{bmatrix} 0 & 1 \\ -3 & 4 \end{bmatrix}$

C $\begin{bmatrix} 0 & 1 \\ -4 & 3 \end{bmatrix}$

D $\begin{bmatrix} 0 & 1 \\ 4 & -3 \end{bmatrix}$

15 MCQ

The matrix of $\mathbf{x}' = A\mathbf{x}$ has eigenvalues -5 and 3. The origin is a:

A stable node

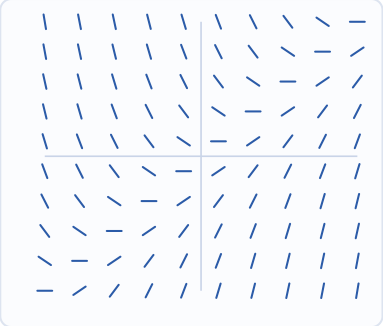
B saddle point (unstable)

C stable spiral

D center (stable, not asymptotically)

16 MCQ

Solve $x' = x,\ y' = 2y$ with $x(0) = 3,\ y(0) = -2$. Find $x(2)$.



A $3e^2$

B $-2e^2$

C $3e^4$

D $-3e^2$

17 MCQ

With $x_1 = y, x_2 = y'$, the equation $y'' - 5y' + y = 0$ becomes $\mathbf{x}' = A\mathbf{x}$. Find A.

A $\begin{bmatrix} 0 & 1 \\ 1 & -5 \end{bmatrix}$

B $\begin{bmatrix} 0 & 1 \\ -1 & 5 \end{bmatrix}$

C $\begin{bmatrix} 1 & 0 \\ -1 & 5 \end{bmatrix}$

D $\begin{bmatrix} 0 & 1 \\ 5 & -1 \end{bmatrix}$

18 MCQ

In this problem, the matrix of $\mathbf{x}' = A\mathbf{x}$ has eigenvalues -3 and 2. The origin is a:

A saddle point (unstable)

B stable spiral

C center (stable, not asymptotically)

D unstable node

19 MCQ

Solve $x' = 3x,\ y' = y$ with $x(0) = -1,\ y(0) = 4$. Find $x(1)$.

A $-e$

B $-e^3$

C e^3

D $4e^3$

20 MCQ

With $x_1 = y, x_2 = y'$, the equation $y'' + 2y = 0$ becomes $\mathbf{x}' = A\mathbf{x}$. Find A.

A $\begin{bmatrix} 0 & 1 \\ 2 & 0 \end{bmatrix}$

B $\begin{bmatrix} 1 & 0 \\ -2 & 0 \end{bmatrix}$

C $\begin{bmatrix} 0 & 1 \\ 0 & -2 \end{bmatrix}$

D $\begin{bmatrix} 0 & 1 \\ -2 & 0 \end{bmatrix}$

21 MCQ

In this problem, the matrix of $\mathbf{x}' = A\mathbf{x}$ has eigenvalues $\pm 4i$. The origin is a:

- **A** stable spiral
- **B** unstable node
- **C** center (stable, not asymptotically)
- **D** stable node

22 MCQ

With $x_1 = y$, $x_2 = y'$, the equation $y'' - 3y' - 2y = 0$ becomes $\mathbf{x}' = A\mathbf{x}$. Find A.

- **A** $\begin{bmatrix} 1 & 0 \\ 2 & 3 \end{bmatrix}$
- **B** $\begin{bmatrix} 0 & 1 \\ 2 & 3 \end{bmatrix}$
- **C** $\begin{bmatrix} 0 & 1 \\ 3 & 2 \end{bmatrix}$
- **D** $\begin{bmatrix} 0 & 1 \\ -2 & -3 \end{bmatrix}$

23 MCQ

The matrix of $\mathbf{x}' = A\mathbf{x}$ has eigenvalues 3 and -4. The origin is a:

- **A** saddle point (unstable)
- **B** unstable spiral
- **C** stable spiral
- **D** center (stable, not asymptotically)

24 MCQ

Solve $x' = -2x,\ y' = -3y$ with $x(0) = -2,\ y(0) = -2$. Find $y(2)$.

- **A** $-\frac{2}{e^4}$
- **B** $\frac{2}{e^6}$
- **C** $-\frac{2}{e^6}$
- **D** $-\frac{4}{e^6}$

25 MCQ

With $x_1 = y$, $x_2 = y'$, the equation $y'' + 4y' + 3y = 0$ becomes $\mathbf{x}' = A\mathbf{x}$. Find A.

- **A** $\begin{bmatrix} 0 & 1 \\ -3 & -4 \end{bmatrix}$
- **B** $\begin{bmatrix} 0 & 1 \\ 3 & 4 \end{bmatrix}$
- **C** $\begin{bmatrix} 1 & 0 \\ -3 & -4 \end{bmatrix}$
- **D** $\begin{bmatrix} 0 & 1 \\ -4 & -3 \end{bmatrix}$